1 **(a)** Hexagon **[B1]**

(b) Obtuse **[B1]**

(c) Equilateral **[B1]**

[Total Marks: 3]

2 **(a)** Chord **[B1]**

(b) Tangent **[B1]**

(c) Sector **[B1]**

(d) Acute **[B1]**

[Total Marks: 4]

3 **(a)** Chord **[B1]**

(b) Tangent **[B1]**

(c) Minor segment (Accept segment) **[B1]**

(d) Isosceles **[B1]**

[Total Marks: 4]

4 **(a)** Radius **[B1]**

(b) Tangent **[B1]**

(c) Diameter **[B1]**

(d) 90 **[B1]**

[Total Marks: 4]

5 **(a)** **(i)** Diameter **[B1]**

(ii) Minor segment (or segment) **[B1]**

(iii) Major sector **[B1]**

(b) Reflex angle (or 260°, reflex) **[B1]**

[Total Marks: 4]

6 **(a)** **(i)** $(n-2) \times 180 = 1080$ or $1080 \div 180$ **[M1]**

8 **[A1]**

(ii) Octagon **[B1]**

(b) Rhombus **[B1]**

(c) Cylinder **[B1]**

[Total Marks: 5]

7 **(a)** **(i)** tangent **[B1]**

(ii) chord **[B1]**

(iii) minor segment (accept segment) **[B1]**

(b) Relationship: perpendicular **[B1]**

Angle $OTQ = 90°$ **[B1]**

[Total Marks: 5]

8 **(a)** Tangent **[B1]**

(b) Radius **[B1]**

(c) 90 **[B1]**

(d) Uses $2 \times \pi \times 5$ **[M1]**

10π **[A1]**

[Total Marks: 5]

9 **(a)** Kite **[B1]**

(b) Sector (or minor sector) **[B1]**

(c) Chord **[B1]**

(d) Congruent **[B1]**

[Total Marks: 4]

10 **(a)** Kite **[B1]**

(b) Perpendicular **[B1]**

(c) Congruent **[B1]**

(d) Obtuse **[B1]**

(e) Perpendicular bisector **[B1]**

[Total Marks: 5]

11 **(a)** $180 - 156 = 24$ **[B1]**

(b) $\frac{360}{24}$ **[M1]**

15 **[A1]**

(c) Decagon **[B1]**

(d) Obtuse **[B1]**

[Total Marks: 5]

12 **(a)** Tangent **[B1]**

(b) Chord **[B1]**

(c) Segment **[B1]**

(d) Sector **[B1]**

(e) Reflex **[B1]**

[Total Marks: 5]

13 **(a)** Isosceles trapezium **[B1]**

(b) Calculates number of sides: $\frac{360^\circ}{36^\circ} = 10$ **[M1]**

Regular decagon (or decagon) **[A1]**

(c) Faces = 12 **[B1]**

Edges = 30 **[B1]**

[Total Marks: 5]

14 **(a)** trapezium **[B1]**

(b) **(i)** obtuse **[B1]**

(ii) $360 - 112 = 248$ **[B1]**

(c) **(i)** 34 **[B1]**

(ii) Angle $BCA = 180^\circ - 112^\circ - 34^\circ = 34^\circ$ (or interior angle $BCD = 68^\circ$) **[M1]**

112 **[A1]**

[Total Marks: 6]

15 **(a)** **(i)** Exterior angle $= 180^\circ - 144^\circ = 36^\circ$ **[M1]**

$\frac{360}{36} = 10$ **[A1]**

(ii) Decagon **[B1]**

(b) **(i)** Sum of interior angles $= (6-2) \times 180^\circ = 720^\circ$ **[B1]**

$x + 2x + (x + 30) + (3x + 60) + 110 + 100 = 720$ OR $7x + 300 = 720$ **[M1]**

$x = 60$ **[A1]**

(ii) Right angle (since $60 + 30 = 90$) **[B1]**

(iii) Reflex angle (since $3(60) + 60 = 240$) **[B1]**

[Total Marks: 8]

16 (a) $\frac{(5-2)\times 180}{5}$ or $180 - \frac{360}{5}$ **[M1]**

108 **[A1]**

(b) $\frac{(6-2)\times 180}{6}$ or $180 - \frac{360}{6}$ **[M1]**

120 **[A1]**

(c) Identifies angles around point A: 108° and 120° **[M1]**

Calculates reflex angle $= 360 - 108 - 120$ **[M1]**

132 **[A1]**

[Total Marks: 7]

17 (a) Similar (geometrically similar) **[B1]**

(b) In a right circular cone, the axis VB is perpendicular to the circular base planes, so $\angle VAC = 90^\circ$ **[B1]**

Both AC and BD lie in parallel planes and are perpendicular to the same line VB in the same cross-sectional plane, so AC is parallel to BD **[B1]**

(c) Uses similarity scale factor: $\frac{VA}{VA+6} = \frac{4.5}{7.5}$ (or $\frac{3}{5}$) **[M1]**

$5VA = 3(VA + 6)$

2VA $= 18[M1]$

9 **[A1]**

[Total Marks: 6]

18 (a) Frustum **[B1]**

(b) (i) Similar **[B1]**

(ii) Scale factor $= \frac{20+10}{10} = 3$ or $\frac{10}{30} = \frac{1}{3}$ **[M1]**

$r = \frac{12}{3} = 4$ **[A1]**

(iii) Radius difference $= 12 - 4 = 8$ **[M1]**

$\sqrt{20^2 + 8^2}$ **[M1]**

21.5 or 21.54... **[A1]**

[Total Marks: 7]

19 (a) Isosceles **[B1]**

(b) $138 + 180$ or $360 - (180 - 138)$ **[M1]**

318 **[A1]**

(c) **(i)** Perpendicular bisector **[B1]**

(ii) Angle $BAC = 138 - 62 = 76°$ **[M1]**

Bearing $= 062 + (76 \div 2) = 100°$ **[A1]**

(d) Trapezium **[B1]**

[Total Marks: 7]

20 **(a)** $\dfrac{h}{h+12} = \dfrac{3}{5}$ **[M1]**

$5h = 3(h + 12)$ **[M1]**

$5h = 3h + 36 \implies 2h = 36 \implies h = 18$ **[A1]**

(b) $\dfrac{r}{r+5} = \dfrac{3}{5}$ or $\dfrac{r}{r+5} = \dfrac{18}{30}$ **[M1]**

$5r = 3r + 15 \implies 2r = 15 \implies r = 7.5$ **[A1]**

(c) Volume ratio (small cone : large cone) $= \left(\dfrac{3}{5}\right)^3 = \dfrac{27}{125}$ **[M1]**

Volume of frustum $= 125 - 27 = 98$, so $k = 98$ **[A1]**

[Total Marks: 7]

21 **(a)** Uses similar triangles to find the height H of the full cone: $\dfrac{H}{12} = \dfrac{H-15}{4}$ **[M1]**

$4H = 12H - 180 \implies 8H = 180 \implies H = 22.5$ cm **[A1]**

Volume of full cone $= \dfrac{1}{3} \times \pi \times 12^2 \times 22.5 = 1080\pi$ **[M1]**

Volume of small cone removed $= \dfrac{1}{3} \times \pi \times 4^2 \times (22.5 - 15) = \dfrac{1}{3} \times \pi \times 16 \times 7.5 = 40\pi$ **[M1]**

Volume of frustum $= 1080\pi - 40\pi = 1040\pi$ or 3267 or 3270 (3 sf) **[A1]**

(b) Identifies the area scale factor is $\dfrac{9}{25}$ **[B1]**

Calculates the length scale factor: $\sqrt{\dfrac{9}{25}} = \dfrac{3}{5}$ **[M1]**

New height $= 15 \times \dfrac{3}{5} = 9$ **[A1]**

[Total Marks: 8]

22 **(a)** Setting up an equation connecting interior and exterior angles, e.g., $4x + x = 180$ **[M1]**

Exterior angle $= 36°$ or Interior angle $= 144°$ **[A1]**

$n = 10$ **[A1]**

Decagon **[B1]**

(b) Faces = 12 **[B1]**

Edges = 30 **[B1]**

Vertices = 20 **[B1]**

(c) Finding the linear scale factor $\sqrt{\frac{16}{9}}$ or $\frac{4}{3}$ **[M1]**

Finding the volume scale factor $\left(\frac{4}{3}\right)^3$ or $\frac{64}{27}$ **[M1]**

256π **[A1]**

[Total Marks: 10]

23 **(a)** Angle ADE = Angle ABC because they are corresponding angles **[B1]**

Angle A is common to both triangles OR Angle AED = Angle ACB (corresponding angles) **[B1]**

(b) Length scale factor $= \frac{6}{10}$ or 0.6 **[M1]**

$DE = 15 \times 0.6 = 9$ **[A1]**

(c) $AC = \frac{7.5}{0.6} = 12.5$ **[M1]**

$EC = 12.5 - 7.5 = 5$ **[A1]**

(d) Trapezium **[B1]**

(e) Area scale factor $= 0.6^2 = 0.36$ **[M1]**

Area of triangle $ABC = \frac{18}{0.36} = 50$ **[M1]**

Area of $DBCE = 50 - 18 = 32$ **[A1]**

[Total Marks: 10]

24 **(a)** **(i)** frustum **[B1]**

(ii) tetrahedron **[B1]**

(b) Reason: the cut is parallel to the base so corresponding angles are equal **[B1]**

Scale factor = 0.4 (or $\frac{2}{5}$) **[B1]**

(c) $10 \times \frac{6}{15}$ **[M1]**

4 **[A1]**

(d) **(i)** Number of sides $= \frac{360}{36} = 10$ **[M1]**

decagon **[A1]**

(ii) Faces = 12 **[B1]**

Edges = 30 **[B1]**

[Total Marks: 10]